MAT367
Intro to Differential Geometry

Class Lecture Notes

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Class Lectures

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

V Vector Fields and Bundles

8 Vector Fields

(check) Lec 13 - Feb 24 (Week 7)

we jump arnd a bit - doing chs 6/9 kinda together.

whats a vector field? morally, a smooth choice of tangent vec at every pt in M. eg, for $M \subseteq \mathbb{R}^n$, tangent vecs live in $T_pM \subseteq \mathbb{R}^n$.

Definition 8.1

a **vector field** is a smooth map $X : M \longrightarrow \mathbb{R}^n$ s.t. $X(p) \in T_pM \subseteq \mathbb{R}^n$.

what does it mean on an abstract mfld? "lets start by cheating". v1: look at charts.

Definition 8.2

a **vector field** X on M is a map $p \mapsto X(p) \in T_pM$, such that for each chart φ, we have $D_p\varphi \circ X : U \longrightarrow \mathbb{R}^n$ is smooth.

this uses the identification $T_pM \simeq \mathbb{R}^n$ induced by chart. but what is that thing between $D_p\varphi \circ X$?

Definition 8.3

the **tangent bundle** TM of M^m is a smooth $2m$-mfld with:

- pt set $\bigsqcup_{p \in M} T_pM$
- (non-max'l) atlas w charts of the form:

$$\left(\bigsqcup_{p \in U} T_pM, (\varphi \circ \pi, D\varphi) \right) \qquad (\varphi \circ \pi, D_p\varphi) : \bigsqcup_{p \in U} T_pM \longrightarrow \mathbb{R}^m \times \mathbb{R}^m$$

for charts (U, φ) in M.

check: π is smooth. remarks:

(1) the tangent bundle comes w a proj $\pi : TM \longrightarrow M$, given by $T_pM \mapsto p$.

(2) for each $p \in M$, the preimg (or fiber) $\pi^{-1}(p)$ is a vspace.

(3) *local triviality:* for each $p \in M$, there's a nbhd $U \ni p$ s.t. the diagram commutes: [TODO: add diagram], where $(D\varphi, \pi)$ is a smooth map and linear on each $\pi^{-1}(q)$.

Lec 15 - Mar 5 (Week 8)

accidentally git yeeted tuesdays notes so those are gone forever

last time: a VF X on M defns a flow $\Phi : J \subseteq \mathbb{R} \times M \longrightarrow M$ where $\dot{\Phi}_t(p) = X_{\Phi_t(p)}$, for some $J \ni \{0\} \times M$ open.

Definition 8.4

a VF is **complete** if $J = \mathbb{R} \times M$.

for instance, $x\partial_x$ is complete w $\Phi_t(x) = e^{tx}$, but $x^2\partial_x$ is not (escapes to ∞ in finite time).

Theorem 8.5

if X has cpt sppt, it is complete.

Proof.

1) if $p \notin \operatorname{supp}(X)$, $X_p = 0$ implies $\Phi_t(p) = p$ for all t. if $p \in \operatorname{supp}(X)$, then $\Phi_t(p) \in \operatorname{supp}(X)$ when defnd.

2) since J open, $\forall \varepsilon > 0$, there is an open $(-\varepsilon, \varepsilon) \times U_\varepsilon$, where:

$$U_\varepsilon \;=\; \{p \in M : \Phi_t(p) \text{ defnd for } t \in (-\varepsilon, \varepsilon)\}$$

these cvr M. by cptness, take finite subcvr $\{U_{\varepsilon_i}\}$. then $U_{\varepsilon_{\min}}$=M.

3) by flow property, $\Phi_{\varepsilon_{\min}/2}$ defnd $\forall p \in M$. thus $\Phi_{n\varepsilon_{\min}/2}(p)$ is defined on $t \geq 0$, and $h < 0$ can be handled the same way.

■

summary: if X a complete VF, its flow defns a 1-param. family of diffeos.

Corollary 8.6

any VF on a cpt mfld is complete.

is the crll out of order? oh well. note:

i) $\Phi_t \circ \Phi_s = \Phi_{t+s}$

ii) $\Phi(t, p)$ is a smooth fn on $\mathbb{R} \times M$

so $\Phi(t, p)$ a smooth fn on $\mathbb{R} \times M$.

Proposition 8.7

if Φ_t satisfies i) and ii) above, its the flow of a VF X, where:

$$X_p(f) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\Phi_t(p))$$

in other words, X_p is the vel vec of $\gamma(t) = \Phi_t(p)$.

Proof.

first, X satisfies the product rule:

$$X_p(fg) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\Phi_t(p))g(\Phi_t(p))$$

its a VF. trust. it is also an integral curve of X, as:

$$\dot{\Phi}_t(p) = \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \Phi_s(\Phi_t(p)) = X_{\Phi_t(p)}$$

sure. ■

example: given $A \in M_n(\mathbb{R})$, let $\Phi_t : \mathbb{R}^m \longrightarrow \mathbb{R}^m$ be:

$$\Phi_t(x) \;=\; e^{tA}x \;=\; \left(\sum_{n=0}^{\infty} \frac{t^n}{n!} A^n\right) x$$

then the corresp. VF is:

$$X(f)(x) \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(e^{tA}x) \;=\; \nabla f \cdot \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} e^{tA}x \;=\; \nabla f \cdot Ax$$

so in "wishy-washy physics notation", we have $X = Ax \cdot \nabla$.

now, an alt defn of the lie bracket. we start with, naturally, the pullback.

Definition 8.8

given $F : M \longrightarrow N$, we denote by $F^* f = f \circ F$ the **pullback** of $f : N \longrightarrow \mathbb{R}$.

if F is a diffeo, then for a VF X on N, we can define $F^* X$ by:

$$F^* X_{F^{-1}(p)} = TF^{-1}(X_p)$$

Definition 8.9

the **lie derivative** of $f : M \longrightarrow \mathbb{R}$ wrt a VF X is given as:

$$L_X f \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi_t^* f \;\in\; C^\infty(M)$$

the lie derivative of a VF Y wrt X is:

$$L_X Y \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi_t^* Y$$

check: $L_X f = X(f)$.

Theorem 8.10

$L_X Y \;=\; [X, Y]$.